

Introduction to Macroeconomics

Spring Semester

<https://chamilo.unige.ch/home/courses/4102016CR/>

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Third Part
Open
Macroeconomics

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graph TD; A[Third Part Open Macroeconomics] --> B[Week 7 Balance of Payments]; A --> C[Week 8 Exchange rates]; A --> D[Week 9 A macroeconomic theory of the open economy];
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Week 7
Balance of Payments

Week 8
Exchange rates

Week 9
**A macroeconomic
theory of the open
economy**

Week 9:
A Macroeconomic Theory of
The Open Economy

We have seen in the previous chapters two different channels through which economies interact: net exports and net capital outflows.

We have also explored a long-run real exchange rate determination model (PPP), but it had no relationship to net exports or net capital outflows, and it was fixed and always equal to 1 (not very informative).

In this chapter, we will see how the trade balance (and, therefore net capital outflows) and the real exchange rate are jointly determined in a general macroeconomic model, where PPP will not fix the real exchange rate.

We will then use this more general model to analyze the impact of an increase in the government's deficit, trade protection, and capital flight on the exchange rate and the trade balance.....

Chapter Contents

- a) A general model of exchange rates and trade balances
- b) Government deficits, exchange rates, and trade balances
- c) Trade policy, exchange rates, and trade balances
- d) Capital flight, exchange rates, and trade balances
- e) Summary

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a. A general model of exchange rates and trade balances

- We are going to take GDP as given, and we are jointly going to explain the determination of exchange rates and trade balances (and therefore net capital outflows)
- We are going to consider two markets: the market for loanable funds and the market for foreign currency exchange

The Market for Loanable Funds

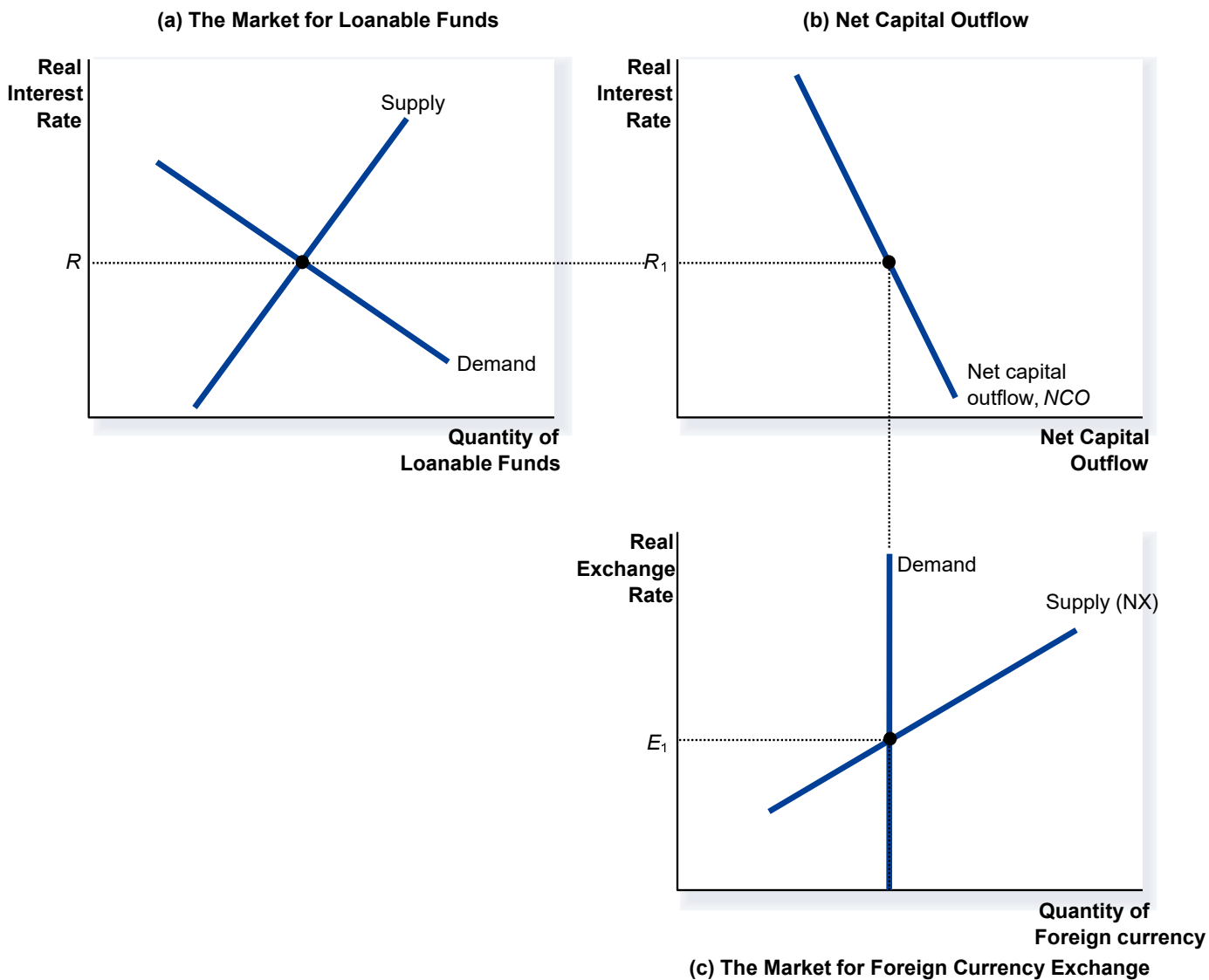
$$S(R) = I(R) + NCO(R)$$

- The supply of loanable funds comes from national savings (S), both private and public.
- The demand for loanable funds comes from domestic investment (I) and net capital outflows (NCO).
- At the equilibrium real interest rate, the amount people want to save exactly balances the desired quantities of investment and net capital outflows.

The market for foreign currency exchange

- In the market for foreign currency exchange, the domestic currency is traded for foreign currencies.
- NCO and NX represent the two sides of the foreign currency exchange market.
- NCO represents the imbalance between the purchases and sales of capital assets and, therefore, the demand for foreign currency.
- NX represents the imbalance between exports and imports of goods and services and therefore the supply of foreign currency.
- For an economy as a whole, NCO and NX must balance each other out, or:
$$NCO(R) = NX(E)$$
- The price that balances the supply and demand for foreign currency is the real exchange rate.
- The supply curve for foreign currency is upward sloping because a higher real exchange rate makes foreign goods more expensive, increasing net exports as it supplies foreign currency to the market.
- The demand curve is vertical because the quantity of foreign currency needed for net capital outflow is unrelated to the real exchange rate.

The joint determination of exchange rates and trade balances



Chapter Contents

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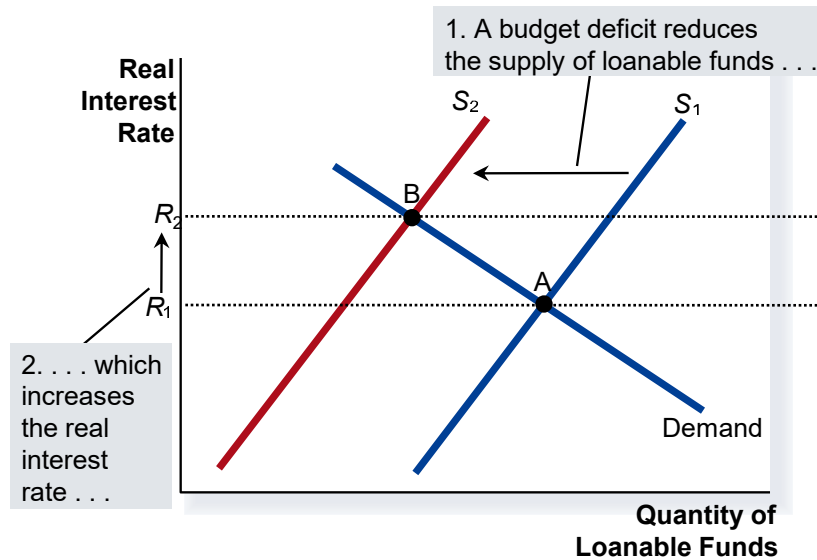
e) Summary

a. Government deficits, exchange rates and trade balances

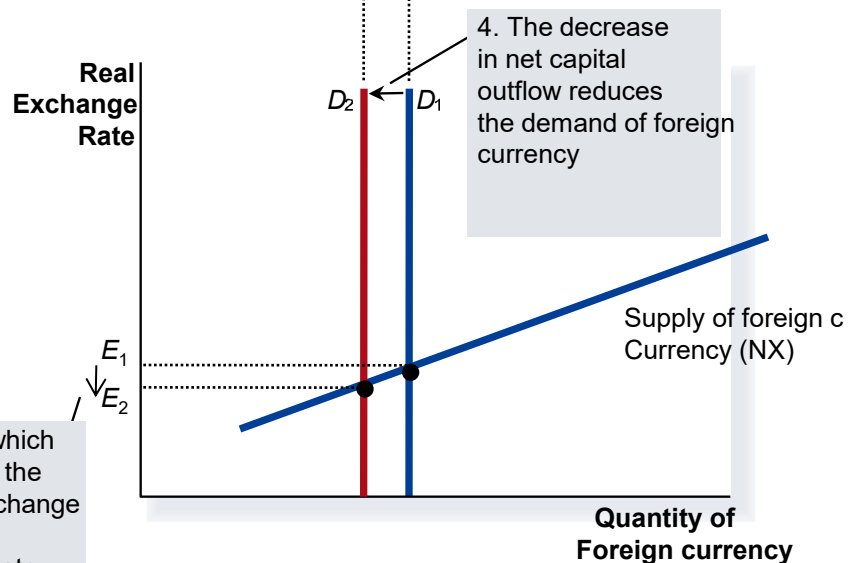
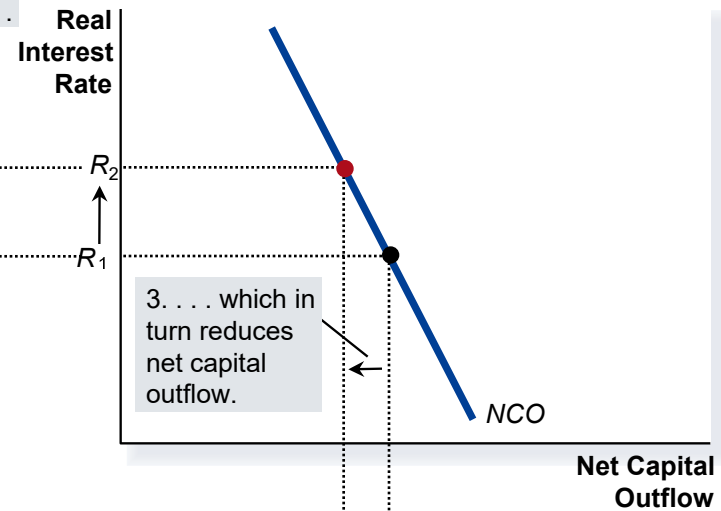
- In an open economy, government budget deficits:
 - reduce the supply of loanable funds (national savings),
 - drive up the interest rate,
 - crowds out domestic investment,
- But this will also have an impact on the trade balance and the exchange rate:
 - To know what happens to the trade balance, let's check what happens with NCO. As the interest rate increases at home, it becomes more interesting to purchase domestic assets (e.g., government bonds) than foreign assets, as the returns on domestic assets are now relatively higher. This will lead to a decline in NCO as everyone prefers now to purchase domestic assets, which will lead to a decline in NX because of the accounting identity at the macroeconomic level.
 - If NCO declines, the demand for foreign currency declines, and this will lead to a lower real exchange rate in equilibrium (an appreciation of the domestic currency)

Impact of a government deficit on the trade balance and the exchange rate

(a) The Market for Loanable Funds



(b) Net Capital Outflow



(c) The Market for Foreign Currency Exchange

The fact that an increase in the trade deficit accompanies an increase in the budget deficit is the reason why we often observe the “twin deficit” phenomenon...(not always as a change in demand for loanable funds or private savings can compensate)

Chapter Contents

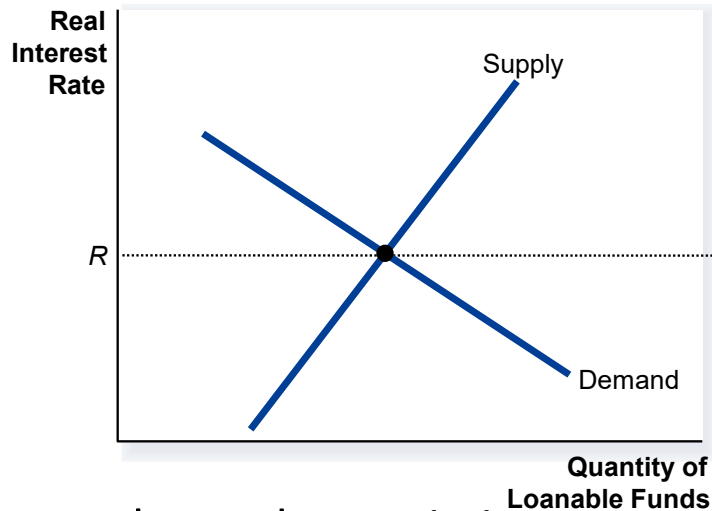
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c. Trade policy, exchange rates and trade balances

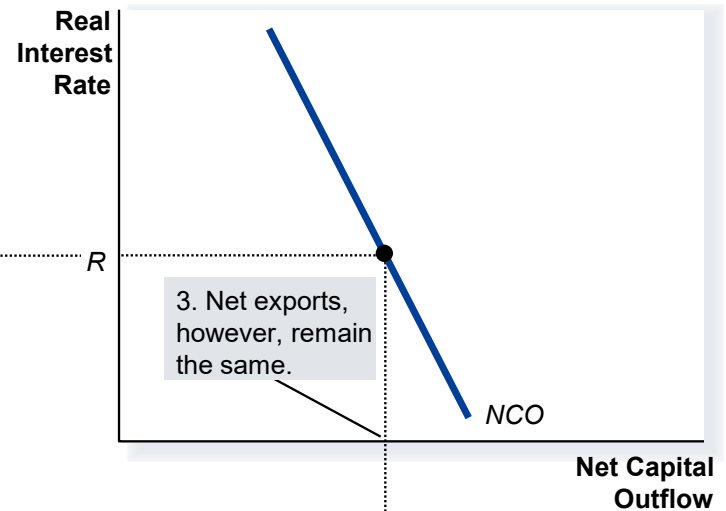
- A *trade policy* is a government policy that directly influences the quantity of goods and services that a country imports or exports.
 - Tariff: A tax on an imported good.
 - Import quota: A limit on the quantity of a good produced abroad and sold domestically.
- Because they do not change national saving or domestic investment, trade policies do not affect NCO and therefore have no impact on the equilibrium trade balance.
 - For a given level of national saving and domestic investment, the real exchange rate adjusts to keep the trade balance the same.
- Trade policies have a greater effect on microeconomic than on macroeconomic markets.
- But a trade restriction will affect the equilibrium exchange rate because it affects net exports for a given exchange rate.

Impact of a restrictive trade policy on the trade balance and the exchange rate

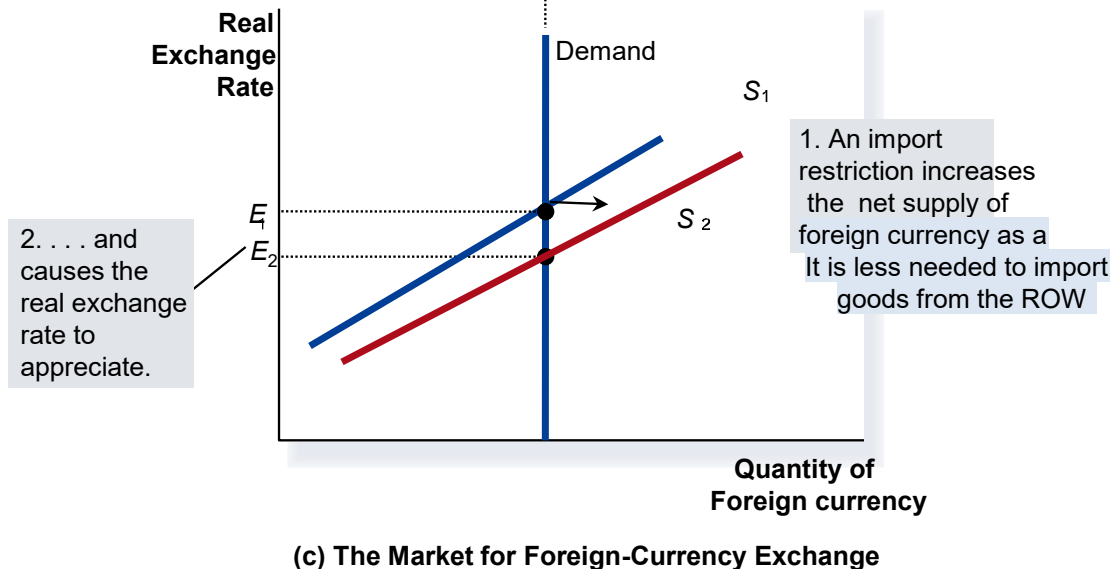
(a) The Market for Loanable Funds



(b) Net Capital Outflow



It is because the trade restriction leads to a reduction in imports and therefore an increase in the net supply of foreign currency, that the exchange rate appreciates, but this hurts exports and helps imports, and this offsets the initial increase in net exports (at the initial exchange rate)



(c) The Market for Foreign-Currency Exchange

Chapter Contents

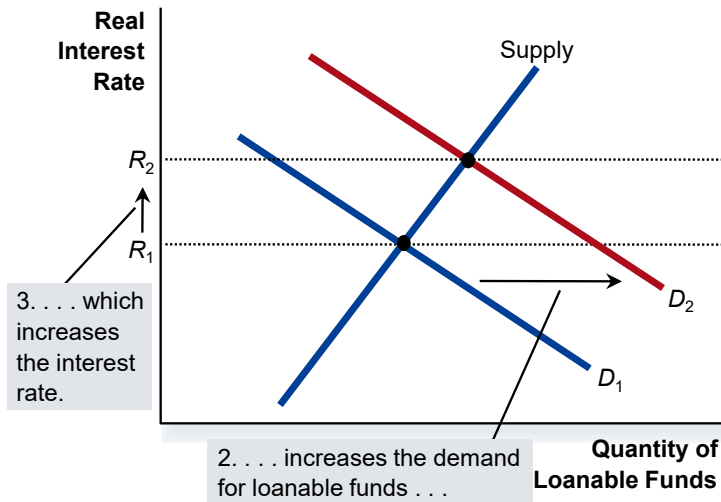
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d. Capital flight, exchange rates and trade balances

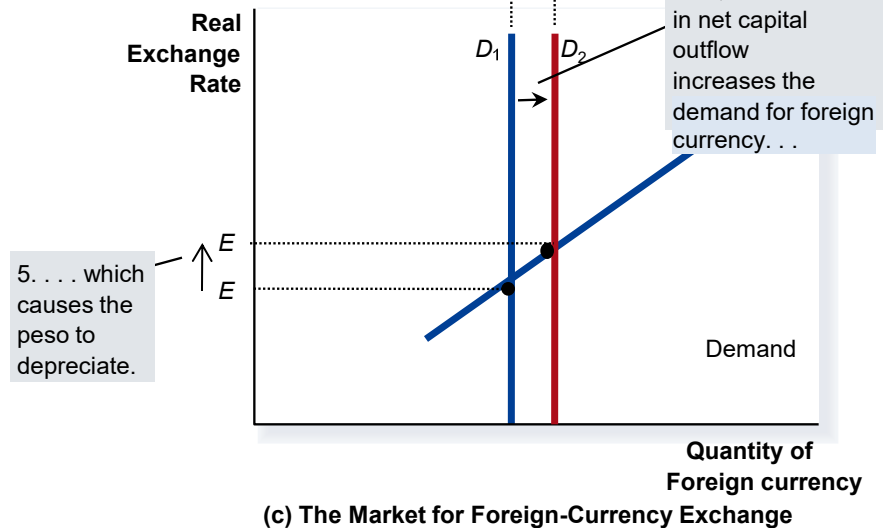
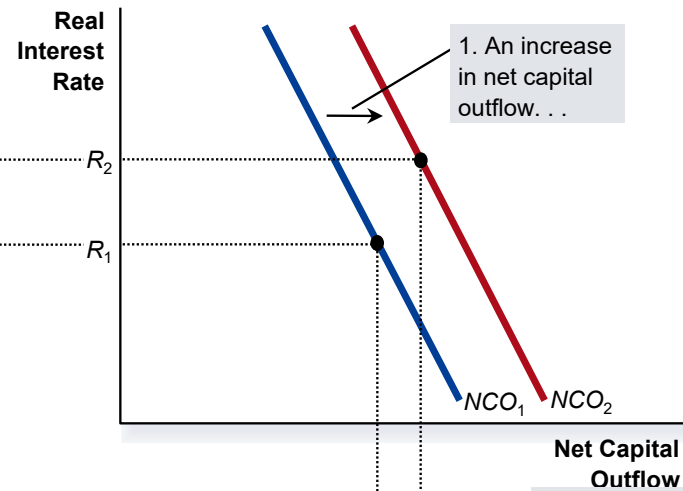
- *Capital flight* is a large and sudden reduction in the demand for assets in a given country.
- If investors become concerned about the safety of their investments, capital can quickly leave an economy.
- Interest rates increase, and the domestic currency depreciates.
- Capital flight has its most significant impact on the country from where the capital is fleeing, but it also affects other countries.
- When investors worldwide observed political problems in Mexico in 1994, they sold some of their Mexican assets and used the proceeds to buy assets from other countries.
- This increased Mexican net capital outflow.
 - The demand for loanable funds increased, increasing the interest rate.
 - This increased the demand for foreign currency in the exchange market.
- The result: a large depreciation of the Mexican peso (“the tequila crisis”)

Impact of a capital flight on the trade balance and the exchange rate

(a) The Market for Loanable Funds in Mexico



(b) Mexican Net Capital Outflow



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e. Summary

- To analyze the macroeconomics of open economies, two markets are central: the market for loanable funds and the market for foreign currency exchange.
- In the market for loanable funds, the interest rate adjusts to balance supply for loanable funds (from national saving) and demand for loanable funds (from domestic investment and net capital outflow).
- In the market for foreign currency exchange, the real exchange rate adjusts to balance the demand for foreign currency (through net capital outflow) and the supply of foreign currency (through net exports).
- Net capital outflow is the variable that connects the two markets.
- A policy that reduces national savings, such as a government budget deficit, reduces the supply of loanable funds and drives the interest rate.
- The higher interest rate reduces net capital outflow, reducing the demand for foreign currency. The domestic currency appreciates (i.e., the foreign currency depreciates), and net exports fall as domestic goods become more expensive.
- This is why we sometimes speak of the twin deficits: budget and trade deficits go hand in hand....

- A trade restriction increases net exports and therefore raises the supply of foreign currency in the market.
- As a result, the domestic currency appreciates (i.e., the foreign currency depreciates), making domestic goods more expensive than foreign goods.
- This appreciation of domestic goods offsets the initial impact of the trade restrictions on net exports, and net exports become again equal to net capital outflows, which were unchanged (determined by the real interest rate, which gets determined in the market for loanable funds).
- When investors change their attitudes about holding assets of a country, the effects on the country's economy can be profound.
- Political instability in a country can lead to capital flight.
- Capital flight tends to increase interest rates as the demand for loanable funds increases when net capital outflows increase.
- If the demand for foreign currency increases, the domestic country depreciates (i.e., foreign currency appreciates).